

**Continuous vs. Discrete - Practice**

Algebra II Honors | Unit 1 | Lesson 3 Practice | TEKS A2.2.A, A2.7.I

**Objective:** I can classify domains and ranges as continuous or discrete, recognize step behavior, and judge when discreteness is a modeling choice.

**Section 1 — Recall**

1. Classify: distance a train travels over a 5-hour trip, given it can go from 0 up to 300 miles. Give the domain.

Classification: \_\_\_\_\_ Domain: \_\_\_\_\_

2. A sequence has domain "the counting numbers." Give the domain in set notation and state whether a sequence is ever continuous.

Domain: { \_\_\_\_\_ } Ever continuous? \_\_\_\_\_

**Section 2 — Apply**

3. A shipping company charges \$8 for a package up to 2 lbs, then \$8 plus \$3 for each additional pound or part of a pound, up to 6 lbs. Complete: 1.0 lb, 2.0 lb, 2.3 lb, 3.8 lb, 5.1 lb. Then classify the domain (weight) and range (cost).

| lbs  | 1.0 | 2.0 | 2.3 | 3.8 | 5.1 |
|------|-----|-----|-----|-----|-----|
| Cost |     |     |     |     |     |

Domain (weight): \_\_\_\_\_ Range (cost): \_\_\_\_\_

4. A vending machine is restocked with 3n snack bars on day n, for n a positive integer. Give the domain in set notation, and explain why a sequence like this is always discrete regardless of context.

Domain: { \_\_\_\_\_ }

\_\_\_\_\_

**Section 3 — Challenge**

5. Explain why a discrete domain can never produce a continuous range.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

6. Is temperature, measured to the nearest 0.1 degree by a digital thermometer, continuous or discrete? Argue both sides, then commit to one, distinguishing the quantity itself from how it is measured.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

**Section 4 — Explain It**

7. A classmate claims any function with a continuous domain must also have a continuous range. Is the classmate correct? Explain, using the step-function idea from today's lesson.

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